Bitcoin Transaction Lab Report

Course: CS216

Topic: Legacy (P2PKH) and SegWit (P2SH-P2WPKH) Transaction Analysis

Student Name: _____

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Part 1: Legacy Transactions (P2PKH)

Workflow

In the legacy transaction experiment, three Bitcoin addresses A, B, and C were created using the P2PKH (Pay-to-Public-Key-Hash) format on the Bitcoin regtest network. 101 blocks were mined to address A so that the mined coins could become spendable. The mined coins created an Unspent Transaction Output (UTXO) which was later used as input for transactions.

Transaction A → B
TXID: 781f6edac495c2afb5744e2ad32c0848e5fae7c227ad78a44a0281eb5474e206
The UTXO belonging to address A was used as the input for a new transaction. The transaction transferred 2.5 BTC to address B and the remaining amount was returned to A as change.

Transaction B → C
TXID: 278b8208ffbf74f46efd000f6ba8a959ca6ece7b12e28b84ab93ae20b5944d29
The output received by B in the previous transaction became a new UTXO. This UTXO was used as the input for the next transaction from B to C.

Decoded Scripts (P2PKH)

Locking Script (scriptPubKey):
76a914c4a7f39f986ecd73cedb63e888692a933150f37f88ac

Decoded Script:
OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG

Unlocking Script (scriptSig):
<signature> <public_key>

Challenge–Response Script Validation

In Bitcoin transactions, the locking script acts as a challenge and the unlocking script acts as a response. The unlocking script pushes the digital signature and public key onto the stack. The locking script then performs operations such as OP_DUP and OP_HASH160 to verify that the provided public key matches the hash stored in the transaction output. Finally, OP_CHECKSIG verifies the digital signature. If all checks succeed, the script evaluates to TRUE and the transaction is considered valid.

Part 2: SegWit Transactions (P2SH- P2WPKH)

Workflow

In the SegWit experiment, three SegWit addresses A', B', and C' were generated using P2SH-P2WPKH format. SegWit separates signature data from the main transaction structure and stores it in the witness field.

Transaction A' → B'
TXID: 575cdd4d17b10bf31799d0ef8f13a8cf392a96ad1959bbe10bdb9d4c7c5e52bf
The UTXO belonging to address A' was used as input. The transaction transferred funds to B' and returned the remaining balance as change.

Transaction B' → C'
TXID: 855ff27ac8717d45534bf3e2c4244ef6be25ecad1c93b7be532de4f9de9cb706
The output received by B' became the input for the next transaction sending funds to address C'.

SegWit Script Structure

Locking Script (scriptPubKey):
OP_HASH160 <script_hash> OP_EQUAL

Redeem Script:
0 <public_key_hash>

Witness Data:
<signature>
<public_key>

In SegWit transactions, the signature and public key are stored in the witness field instead of the scriptSig field.

Part 3: Analysis and Comparison

Transaction Size Comparison

P2PKH transactions store the signature and public key directly inside the scriptSig. In SegWit transactions, the signature data is moved to the witness structure.

Example comparison:
P2PKH Transaction: size ≈ 222 bytes, weight ≈ 888
P2SH-P2WPKH Transaction: size ≈ 223 bytes, vsize ≈ 142, weight ≈ 565

Script Structure Comparison

P2PKH Script Structure:
scriptSig: <signature> <public_key>
scriptPubKey: OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG

P2SH-P2WPKH Script Structure:
scriptSig: redeem script
scriptPubKey: OP_HASH160 <script_hash> OP_EQUAL
witness: <signature> <public_key>

Why SegWit Transactions are Smaller

SegWit separates signature data from the main transaction. Witness data is discounted when calculating block weight. Because of this, SegWit transactions have a smaller effective virtual size (vsize) compared to legacy transactions.

Benefits of SegWit

1. Reduced transaction size
2. Lower transaction fees
3. Increased block capacity
4. Fixes transaction malleability
5. Enables advanced scaling solutions such as the Lightning Network

Screenshots to Include

Insert screenshots of the following:
• Decoded transaction A → B
• Decoded transaction B → C
• ScriptSig and ScriptPubKey outputs
• Bitcoin debugger execution
• SegWit decoded transactions
• Witness data validation